

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

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Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-85050

The expression $3x + 7 + 8x + 11$ can also be written as

$10x + 19$ ☐ $11x + 18$ ☐ $18x + 11$ ☐ $29x$ ☐

2. Number and Algebra, NAP-32096

If $x = 5$, what is the value of $\frac{3x}{2x - 5}$?

$\frac{2}{3}$ ☐ $\frac{3}{4}$ ☐ $\frac{4}{15}$ ☐

3. Number and Algebra, NAP-08602

The value of y can be calculated by using the rule $y = 8 - x^2$.

What is the value of y when $x = 2.5$?

1.75 ☐ 3 ☐ 5.5 ☐ 6.25 ☐

4. Number and Algebra, NAP-02659

The distance between the Earth and the moon is 384 712 303 metres.

How far is that to the nearest million?

☐ 380 000 000
☐ 385 000 000
☐ 384 700 000
☐ 384 710 000

5. Number and Algebra, NAP-32124

Type II Diabetes in America				
Year	2007	2008	2009	2010
Number of people (millions)	31.16	32.42	32.93	33.21

Between 2007 and 2010, the number of people with type II diabetes in America increased by about

☐ 0.5 million people.
☐ 1 million people.
☐ 1.5 million people.
☐ 2 million people.

6. Number and Algebra, NAP-08603

Jack was given a question without a calculator.

$$22 - 16 \times \sqrt{8.5 + 7.5} =$$

Which operation should Jack perform first?

$22 - 16$ ☐ $16 \times \sqrt{8.5}$ ☐ $\sqrt{8.5}$ ☐ $8.5 + 7.5$ ☐

7. Number and Algebra, NAP-85108

The table below gives the lengths of two similar fleas.

Flea	Length
Pulex Flea	$\frac{1}{8}$ inch
Cat Flea	4 millimetres

One inch equals **25.4** millimetres.

What is the difference in millimetres between the length of a Cat Flea and a Pulex Flea?

Give your answer to three decimal places.

 millimetres

8. Number and Algebra, NAP-08635

What is the value of $6 + 2x - x^2$ when $x = -3$?

−93921

☐☐☐☐

Worked Solutions

1. Number and Algebra, NAP-85050

$$3x + 7 + 8x + 11 = 11x + 18$$

2. Number and Algebra, NAP-32096

If $x = 5$,

$$\begin{aligned} \frac{3x}{2x-5} &= \frac{3 \times 5}{(2 \times 5) - 5} \\ &= \frac{15}{3} \\ &= 3 \end{aligned}$$

3. Number and Algebra, NAP-08602

$$\begin{aligned} y &= 8 - x^2 \\ &= 8 - (2.5)^2 \\ &= 8 - 6.25 \\ &= 1.75 \end{aligned}$$

4. Number and Algebra, NAP-02659

$$385\,000\,000$$

5. Number and Algebra, NAP-32124

$$33.21 - 31.16 = 2.05$$

∴ The number increased by about 2 million people.

6. Number and Algebra, NAP-08603

The sum of the numbers under the square root symbol need to be calculated first.

∴ The first operation is $8.5 + 7.5$.

7. Number and Algebra, NAP-85108

$$\frac{1}{8} \times 25.4 = 3.175$$

$$4 - 3.175 = 0.825 \text{ mm}$$

8. Number and Algebra, NAP-08635

$$6 + 2x - x^2 = 6 + 2(-3) - (-3)^2$$

$$= 6 - 6 - 9$$

$$= -9$$